n cooled CSTRs in series: 3^n steady states, 2^n of them stable

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Abstract. In his 1987 thesis on two continuous stirred tank reactors (CSTRs) in series with a first-order exothermic reaction, each with its own cooling jacket, B. Y. Bisbis showed that two reactors can have nine steady states and conjectured that n reactors have at most 3^n steady states, at most 2^n of them stable. We prove the conjecture for the model of the thesis, with Arrhenius kinetics and $\varepsilon = RT_0/E > 0$. The upper bounds hold for all parameter values, and also in the exponential limit $\varepsilon = 0$: one reactor has at most three steady states for any feed, the Jacobian of the cascade is block lower triangular, and the middle state of a reactor with three states is a saddle. The bounds are attained for every n and every $\varepsilon > 0$. For n identical reactors (equal volumes, equal jackets, equal coolant temperature) we give explicit parameter values with exactly 3^n steady states, all hyperbolic, of which exactly 2^n are asymptotically stable. The construction uses the bounded Arrhenius factor. The Damköhler number at infinite temperature is $k_0V/F = 1/(4n)$, so a reactor converts less than $1/(4n+1)$ of the reactant it receives, even when ignited, and at least $e^{-1/4}$ of the feed reaches the last reactor. Strong cooling makes each reactor almost insensitive to the temperature of its feed. The proof reduces to five elementary inequalities, checked in ball arithmetic. Interval certificates (Krawczyk test on the original equations) confirm the 3^n states and the 2^n stable ones for $n \le 6$. Two certified one-reactor examples show that the outer states need not be stable: three states with an unstable upper focus, and three states that are all unstable. The parameters that attain 3^n are far from practical operation: ignited reactors run above $3E/R$. The exponential limit $\varepsilon = 0$, which the thesis uses in its numerical example, is not covered by the construction.

The question

B. Y. Bisbis, *On the steady states and dynamic behavior of two CSTR's in series*, PhD thesis, University of Missouri–Columbia, December 1987, supervisor P. C.-H. Chan [1]. We read the thesis in full (open access). Chapter II proves, with the sequential singularity theory of Dangelmayr and Stewart [2] [3], that two reactors can have nine steady states, and gives a numerical example with $\varepsilon = 0$. The concluding remarks of Chapter II (p. 27) state:

> *"For the case of n - CSTR's in series, one may conjecture that reiteration of the same methods used here, along with an induction reasoning, would lead to the following : The maximum possible number of steady states is* 3^n*, of which at most only* 2^n *are stable, the rest unstable."*

The abstract repeats that "some global statements about maximum multiplicity in the case of n - CSTR's in series are made". On the parameters (p. 29): "we have the same reaction taking place in both reactors and therefore kinetic parameters such as B, Da_0 and ε are chosen only once. [...] The rest of the parameters α, η_1, η_2, β_1 and β_2 may be chosen independently". The residence time τ is also common, since one flow rate passes through all reactors.

The model. The thesis writes the balances of reactor i with volume V_i, jacket coefficient u_ia_i and coolant temperature T_{ci} (eqs. (1)–(4)) and the dimensionless form (eqs. (5)–(8),

p. 4; the scaling of Uppal, Ray and Poore [4]) with $\theta_i = \frac{E}{RT_0}\frac{T_i - T_0}{T_0}$, $y_i = (c_{A0} - c_{Ai})/c_{A0}$, $\varepsilon = RT_0/E$, $B = (-\Delta H)c_{A0}/(\varepsilon \rho C_p T_0)$, $\eta_i = \frac{E}{RT_0}\frac{T_{ci} - T_0}{T_0}$, $\mathrm{Da}_0 = k_0 e^{-1/\varepsilon} t_r$, $\tau = V_1/(F t_r)$, $\alpha = V_2/V_1$. The n-reactor version is, for $k = 1, \dots, n$,

$$\begin{aligned} \frac{dy_k}{dt} &= \frac{y_{k-1} - y_k}{\alpha_k} + \mathrm{Da}_0 \tau (1 - y_k) E(\theta_k), \\ \frac{d\theta_k}{dt} &= \frac{\theta_{k-1} - \theta_k}{\alpha_k} + \mathrm{Da}_0 \tau B (1 - y_k) E(\theta_k) - \frac{\beta_k \tau}{\alpha_k}(\theta_k - \eta_k), \end{aligned} \qquad E(\theta) = \exp\Big(\frac{\theta}{1 + \varepsilon\theta}\Big), \tag{1}$$

with $y_0 = \theta_0 = 0$ (fresh feed at T_0), $\alpha_1 = 1$ and $\alpha_k = V_k/V_1$. For $n = 2$ this is eqs. (5)–(8) of the thesis verbatim. The parameters are $\varepsilon > 0$, $B > 0$, $\mathrm{Da}_0 > 0$, $\tau > 0$ (common), and $\alpha_k > 0$, $\beta_k \geq 0$, $\eta_k > -1/\varepsilon$ (per reactor; the last condition says $T_{ck} > 0$). The phase space is $\{(y_k, \theta_k)_{k \leq n} : \ 1 + \varepsilon\theta_k > 0\}$, that is, positive absolute temperatures. A steady state is an equilibrium of (1). *Stable* means stable in the sense of Lyapunov. For the upper bounds we also allow $\varepsilon = 0$, $E(\theta) = e^{\theta}$, $\theta_k, \eta_k \in \mathbb{R}$ (the Frank-Kamenetskii approximation, used in the numerical example of the thesis).

Remark 2 (on β_k)**.** The thesis lists $\beta_i = u_i a_i t_r/(V_i \rho C_p)$, whereas eq. (8) as printed corresponds to normalising by V_1. With the other convention, $\beta_k \tau/\alpha_k$ becomes $\beta_k \tau$ in (1). Nothing below depends on this: in Theorem 9 only the number $b_k \geq 0$ defined in (4) matters (it is $\alpha_k \beta_k \tau$ with the other convention), and the construction of Theorem 14 has $\alpha_k = 1$, where both conventions agree. ⌟

Prior work. Varma [5] gave the bound $2^{n+1} - 1$ steady states, at most $n + 1$ stable, for n reactors in series, "for the adiabatic and the isothermal occurrence of the reaction" according to [1] (see also [6]). In those cases each reactor has a single state variable. We have not read [5], which is paywalled. Svoronos, Aris and Stephanopoulos [7] studied two identical tanks in the infinite-activation-energy approximation ($\varepsilon = 0$, according to [1]); their abstract reports up to seven steady states for a given Damköhler number, at most three of them stable. Kubíček et al. [8] studied two nonadiabatic reactors numerically. Bisbis [1] and Retzloff, Chan, Chicone and Papick [9] proved that nine states occur for two jacketed reactors. Li [10] treats isothermal reactions in two CSTRs; his abstract (paraphrased in secondary sources) extends the count to 3^N states, 2^N stable, for N isothermal reactors. That is a different model, without energy balance. We found no proof, for any $n \geq 3$, that 3^n states are attained in the jacketed nonisothermal model.

Theorem 3 (main result)**. (1)** For every $n \geq 1$ and all parameter values ($\varepsilon \geq 0$), system (1) has at most 3^n steady states, and at most 2^n of them are stable (Theorem 9).

(2) For every $n \geq 1$ and every $\varepsilon > 0$, put $m = \max\{1, \varepsilon\}$ and

$$B = \frac{6400\, n\, m}{\varepsilon^2}, \qquad \mathrm{Da}_0 = \frac{e^{-1/\varepsilon}}{4n}, \qquad \tau = 1,$$
$$\alpha_k = 1, \qquad \beta_k = 200m - 1, \qquad \eta_k = \frac{1 - 20\varepsilon}{20\varepsilon^2} \qquad (k = 1, \dots, n).$$

Then (1) has exactly 3^n steady states. All are hyperbolic; exactly 2^n are asymptotically stable, and the other $3^n - 2^n$ are unstable, each with a positive real eigenvalue (Theorem 14).

(3) Outer steady states need not be stable: there are parameters with $n = 1$ and three steady states of which only one is stable, and parameters with three steady states none of which is stable (Proposition 11).

In physical units, the family of part 2 has identical reactors with $k_0V/F = 1/(4n)$, $UA/(F\rho C_p) = 200m - 1$, coolant temperature $T_c = E/(20R)$ and adiabatic temperature rise $(-\Delta H)c_{A0}/(\rho C_p) = 6400\, n\, m\, E/R$. The feed temperature $T_0 = \varepsilon E/R$ is arbitrary. For $\varepsilon = 1/20$ the coolant is at the feed temperature ($\eta_k = 0$).

1 One reactor

Fix a reactor k and its inlet (y_{k-1}, θ_{k-1}), and write $c_{k-1} = 1 - y_{k-1}$. Set

$$D_k = \alpha_k \mathrm{Da}_0 \tau, \quad b_k = \beta_k \tau, \quad \lambda_k = \frac{1}{1+b_k} \in (0,1], \quad g_k(\theta) = \frac{D_k E(\theta)}{1 + D_k E(\theta)}, \tag{4}$$

$$\varphi_k(\theta) = \lambda_k \theta_{k-1} + (1-\lambda_k)\eta_k + \lambda_k B c_{k-1}\, g_k(\theta) - \theta. \tag{5}$$

Lemma 6 (reduction). Let $c_{k-1} > 0$. The pair (y_k, θ_k) makes the right-hand side of reactor k in (1) vanish if and only if $\varphi_k(\theta_k) = 0$ and $1 - y_k = c_{k-1}/(1 + D_k E(\theta_k))$. In particular $0 < 1 - y_k \le c_{k-1}$ at a steady state, and the steady states of (1) are the sequences obtained by choosing, for $k = 1, \dots, n$ in turn, a zero θ_k of φ_k (which depends on the choices for reactors $1, \dots, k-1$).

Proof. Multiply the first equation by α_k: it vanishes iff $c_{k-1} - c_k = D_k c_k E(\theta_k)$ with $c_k = 1 - y_k$, that is $c_k = c_{k-1}/(1 + D_k E)$. On this curve, α_k times the second equation equals $(1 + b_k)\varphi_k(\theta_k)$ (check A2 of `check_model_algebra.py`). Since $c_0 = 1$, induction gives $c_k > 0$ for all k. □

Lemma 7 (at most three states). Let $B > 0$, $c_{k-1} > 0$, $\varepsilon \ge 0$. Then φ_k has at most three zeros on $I = \{\theta : 1 + \varepsilon\theta > 0\}$. If it has three zeros $\theta^1 < \theta^2 < \theta^3$, then $\varphi_k'(\theta^1) < 0$, $\varphi_k'(\theta^2) > 0$ and $\varphi_k'(\theta^3) < 0$.

Proof. Write $g = g_k$. A direct computation (checks A4) gives

$$g' = \frac{g(1-g)}{(1+\varepsilon\theta)^2} > 0, \qquad g'' = \frac{g(1-g)}{(1+\varepsilon\theta)^4}\, s(\theta), \qquad s(\theta) = 1 - 2g(\theta) - 2\varepsilon(1+\varepsilon\theta).$$

The function s is strictly decreasing on I, because g is strictly increasing and $\varepsilon(1+\varepsilon\theta)$ is nondecreasing. So $g'' > 0$ on an interval I_- (possibly empty) and $g'' < 0$ on the complementary interval I_+ to its right, up to at most one point. Since $\varphi_k'' = \lambda_k B c_{k-1} g''$, the derivative φ_k' is strictly increasing on I_- and strictly decreasing on I_+. It has at most two zeros, and by Rolle's theorem φ_k has at most three.

Suppose there are three zeros. By Rolle, φ_k' vanishes at some $\xi_1 \in (\theta^1, \theta^2)$ and some $\xi_2 \in (\theta^2, \theta^3)$. These are its only zeros, so $\varphi_k'(\theta^j) \ne 0$ and φ_k changes sign at each θ^j. Since $g < 1$, $\varphi_k(\theta) \to -\infty$ as $\theta \to \infty$. Hence $\varphi_k < 0$ on (θ^3, ∞), $\varphi_k > 0$ on (θ^2, θ^3) and $\varphi_k < 0$ on (θ^1, θ^2), and the signs of $\varphi_k'(\theta^j)$ follow. □

Lemma 8 (linearisation). Order the variables as $(y_1, \theta_1, \dots, y_n, \theta_n)$. The Jacobian of (1) is block lower triangular with 2×2 diagonal blocks J_k; the block of reactor k depends only on (y_k, θ_k) and the parameters. At a steady state, with $\mathcal{E}_k = D_k E(\theta_k)$,

$$\alpha_k^2 \det J_k = -(1+\mathcal{E}_k)(1+b_k)\,\varphi_k'(\theta_k),$$
$$\alpha_k \operatorname{tr} J_k = (1+\mathcal{E}_k)(1+b_k)\Big(\varphi_k'(\theta_k) - \lambda_k + \frac{\mathcal{E}_k}{1+\mathcal{E}_k}\Big).$$

Hence the eigenvalues of the Jacobian are those of the blocks, and

(1) if $\varphi_k'(\theta_k) > 0$, then J_k has a positive and a negative real eigenvalue;

(2) J_k is Hurwitz (both eigenvalues with negative real part) if and only if $\varphi_k'(\theta_k) < \min\{0,\ \lambda_k - \mathcal{E}_k/(1+\mathcal{E}_k)\}$.

Proof. Reactor k involves only $(y_{k-1}, \theta_{k-1}, y_k, \theta_k)$, which gives the block structure (check A7 for $n = 3$). The two identities are checks A3. A real 2×2 matrix is Hurwitz iff its determinant is positive and its trace negative, and has real eigenvalues of opposite signs iff its determinant is negative. □

2 Upper bounds

Theorem 9 (upper bounds). Let $n \geq 1$, $\varepsilon \geq 0$, $B, \mathrm{Da}_0, \tau, \alpha_k > 0$, $\beta_k \geq 0$, $\eta_k \in \mathbb{R}$.

(1) System (1) has at most 3^n steady states.

(2) A steady state at which $\varphi_k'(\theta_k) > 0$ for some k has a positive real eigenvalue, so it is unstable. At most 2^n steady states satisfy $\varphi_k'(\theta_k) \leq 0$ for all k. Hence at most 2^n steady states are stable.

(3) If there are exactly 3^n steady states, then exactly $3^n - 2^n$ of them have a reactor with $\varphi_k'(\theta_k) > 0$, and these are unstable.

Proof. By Lemma 6, the steady states form a rooted tree of depth n. A node at depth $k-1$ is a steady state of reactors $1, \dots, k-1$; its children are the zeros of the corresponding φ_k. By Lemma 7 every node has at most three children, so there are at most 3^n leaves. This proves part 1.

If $\varphi_k'(\theta_k) > 0$, then J_k has a positive eigenvalue (Lemma 8), hence so has the full Jacobian, and a steady state at which the Jacobian has an eigenvalue with positive real part is unstable (Lyapunov; see §2.9 [11]). Every node has at most two children with $\varphi_k' \leq 0$: if it has three children, the middle one has $\varphi_k' > 0$ by Lemma 7. So at most 2^n leaves have $\varphi_k' \leq 0$ along the whole path, which proves part 2. If there are 3^n leaves, every node has exactly three children, exactly two of them with $\varphi_k' < 0$, and exactly 2^n paths avoid the middle child; this proves part 3. □

Remark 10. The upper bound uses only that each reactor has at most three steady states for every feed, and the triangular structure. The same proof applies if the reactors ⤵

differ in any per-reactor parameter, and for $\varepsilon = 0$.

Proposition 11 (outer states can be unstable). Take $n = 1$, $\varepsilon = 1/20$, $\tau = 1$.

(1) For $B = 20$, $\mathrm{Da}_0 = 1/10$, $\beta_1 = 2$, $\eta_1 = -1$ there are exactly three steady states, $\theta \approx -0.1279$, 2.3968, 4.9257: a hyperbolic sink, a saddle, and an unstable focus (trace ≈ 1.596, determinant ≈ 7.815).

(2) For $B = 30$, $\mathrm{Da}_0 = 69/500$, $\beta_1 = 5$, $\eta_1 = -7/20$ there are exactly three steady states, $\theta \approx 1.3456$, 1.7744, 2.6969: an unstable node, a saddle, and an unstable node. No steady state is stable.

Proof. Computer-assisted (`certify_unstable_outer_state.py`). For each example, ball arithmetic shows three sign changes of φ_1 on a grid of step $1/32$, so by Lemma 7 there are exactly three zeros. Each steady state is then verified by a Krawczyk test in (y, θ), and the trace, determinant and discriminant of J_1 are enclosed on the verified box; their signs give the stated types. The enclosures are in `certificates/unstable_outer_state.json`. □

So the two outer states of a reactor can both be unstable. The bound 2^n is therefore not reached merely because there are 3^n steady states. Theorem 14 shows that it is reached.

3 Attainment for every n

From now on $\varepsilon > 0$. We use the reduced temperature $z = RT/E = \varepsilon(1 + \varepsilon\theta) > 0$, so that $E(\theta) = e^{1/\varepsilon} e^{-1/z}$, and the reduced concentration $c = 1 - y$. For reactor k set

$$K_k = D_k e^{1/\varepsilon} = \frac{k_0 V_k}{F}, \quad \Delta = \varepsilon^2 B = \frac{R\,(-\Delta H) c_{A0}}{E \rho C_p}, \quad z_{c,k} = \varepsilon(1 + \varepsilon\eta_k) = \frac{RT_{ck}}{E} > 0.$$

K_k is the Damköhler number at infinite temperature and Δ the adiabatic temperature rise in units of E/R. The inlet of reactor k is (c_{k-1}, z_{k-1}), with $(c_0, z_0) = (1, \varepsilon)$.

Lemma 12 (reduced variables). With $k(z) = K_k e^{-1/z}$ and

$$F_k(z) = \lambda_k z_{k-1} + (1 - \lambda_k) z_{c,k} + \lambda_k \Delta c_{k-1} \frac{k(z)}{1 + k(z)} - z,$$

one has $F_k(z) = \varepsilon^2 \varphi_k(\theta)$ and $F_k'(z) = \varphi_k'(\theta)$ for $z = \varepsilon(1 + \varepsilon\theta)$. The outlet of a steady state of reactor k is $c_k = c_{k-1}/(1 + k(z_k))$. Moreover, with $X_k = \lambda_k \Delta K_k$ and $\psi(z) = e^{-1/z}/z^2$,

$$F_k'(z) = X_k c_{k-1} \frac{\psi(z)}{(1 + k(z))^2} - 1.$$

Proof. Substitution (checks A1 and A5). □

Note that $\psi'(z) = e^{-1/z} z^{-4}(1 - 2z)$ (check A6). So ψ increases on $(0, 1/2]$ and decreases on $[1/2, \infty)$. Also $k/(1+k) < K_k/(1+K_k)$ and $e^{-1/z}/(1 + K_k e^{-1/z}) = 1/(e^{1/z} + K_k) < 1/(1 + K_k)$ for all $z > 0$.

Lemma 13 (window lemma). Fix a reactor with parameters $\lambda \in (0,1)$, $K > 0$, $z_c > 0$, $\Delta > 0$, and put $X = \lambda\Delta K$. Let $0 < a < 1/2 < b$, $0 < c_- \le 1$ and $\zeta_+ > 0$ satisfy

$$\lambda\zeta_+ + z_c + Xe^{-1/a} < a, \tag{W1}$$
$$Xc_-e^{-1/b} > (1+K)\,b, \tag{W2}$$
$$(1+K)X\psi(a) < 1 \quad \text{and} \quad (1+K)X\psi(b) < 1. \tag{W3}$$

Then for every inlet (c,ζ) with $c_- \le c \le 1$ and $0 < \zeta \le \zeta_+$:

(1) the reactor has exactly three steady states, with temperatures $z_L \in (0,a)$, $z_M \in (a,b)$ and $z_H \in (b,\infty)$;

(2) at z_L and z_H the block J is Hurwitz; at z_M it has real eigenvalues of opposite signs;

(3) every outlet satisfies $c_-/(1+K) \le c/(1+K) < c_{\text{out}} < c \le 1$ and $0 < z_{\text{out}} < \lambda\zeta_+ + (1-\lambda)z_c + X/(1+K)$.

Proof. Write $F(z) = \lambda\zeta + (1-\lambda)z_c + Xc\,e^{-1/z}/(1+Ke^{-1/z}) - z$, which is F_k of Lemma 12 (since $\lambda\Delta c\,k/(1+k) = Xc\,e^{-1/z}/(1+Ke^{-1/z})$).

Signs. As $z \to 0^+$, $F(z) \to \lambda\zeta + (1-\lambda)z_c > 0$. At $z = a$, using $c \le 1$, $\zeta \le \zeta_+$ and $(1-\lambda)z_c \le z_c$, $F(a) \le \lambda\zeta_+ + z_c + Xe^{-1/a} - a < 0$ by (W1). At $z = b$, using $c \ge c_-$ and $1 + Ke^{-1/b} < 1 + K$, $F(b) > Xc_-e^{-1/b}/(1+K) - b > 0$ by (W2). For $z \ge z^* := \lambda\zeta_+ + (1-\lambda)z_c + X/(1+K)$, $F(z) < z^* - z \le 0$. So F has a zero in each of $(0,a)$, (a,b) and (b,z^*). By Lemma 7 and Lemma 12 there are no others. This proves part 1 and the bound on z_{out}.

Stability. At $z_L < a < 1/2$, $\psi(z_L) < \psi(a)$; at $z_H > b > 1/2$, $\psi(z_H) < \psi(b)$. For $z \in \{z_L, z_H\}$, Lemma 12 and (W3) give

$$F'(z) + 1 = Xc\,\frac{\psi(z)}{(1+k)^2} \le X\psi(z) < \frac{1}{1+K} < \frac{1}{1+k}\,.$$

Hence $F'(z) < -k/(1+k) < \lambda - k/(1+k)$ and $F'(z) < 0$. By Lemma 8(2) (with $\mathcal{E} = k$ and $\varphi' = F'$) the block is Hurwitz. At z_M, $F'(z_M) > 0$ by Lemma 7, and Lemma 8(1) applies.

Outlet concentration. $c_{\text{out}} = c/(1+k)$ with $0 < k < K$. □

Theorem 14 (cascade design). Let $n \ge 1$ and $\varepsilon > 0$. Suppose all n reactors have the same $\lambda \in (0,1)$, $K > 0$, $z_c > 0$ (and $\Delta > 0$ is common). Suppose that (W1)–(W3) hold with

$$c_- = (1+K)^{-(n-1)}, \qquad \zeta_+ \ge \max\Big\{\varepsilon,\ \frac{z_c + X/(1+K)}{1-\lambda}\Big\}.$$

Then (1) has exactly 3^n steady states, labelled by the words $w \in \{L,M,H\}^n$, where the letter w_k says in which window of Lemma 13 the temperature of reactor k lies. All are hyperbolic. The 2^n states with $w \in \{L,H\}^n$ are asymptotically stable. The

other $3^n - 2^n$ states have a positive real eigenvalue and are unstable. In particular this holds for the parameters of Theorem 3(2).

Proof. *Induction.* We show, for $k = 0, \dots, n$, that the steady states of the subsystem formed by reactors $1, \dots, k$ are in bijection with $\{L, M, H\}^k$, and that every outlet satisfies $(1+K)^{-k} \le c_k \le 1$ and $0 < z_k \le \zeta_+$. For $k = 0$ the only "outlet" is the feed $(1, \varepsilon)$, and $\varepsilon \le \zeta_+$. Suppose the claim holds for some $k < n$. Every inlet (c_k, z_k) of reactor $k+1$ has $c_k \ge (1+K)^{-k} \ge c_-$ and $0 < z_k \le \zeta_+$. By Lemma 13 reactor $k+1$ has exactly three steady states, one in each window, so each word of length k extends in exactly three ways. The new outlets satisfy $c_{k+1} \ge c_k/(1+K) \ge (1+K)^{-k-1}$ and

$$z_{k+1} < \lambda\zeta_+ + (1-\lambda)z_c + \frac{X}{1+K} \le \lambda\zeta_+ + (1-\lambda)\zeta_+ = \zeta_+.$$

Distinct words give distinct states: at the first letter where two words differ, the temperatures of that reactor lie in disjoint windows. By Lemma 6 every steady state arises in this way.

Stability. By Lemma 8 the spectrum of the Jacobian is the union of the spectra of the blocks J_k. By Lemma 13(2), every block is Hurwitz when $w_k \in \{L, H\}$, and has real eigenvalues of opposite signs when $w_k = M$. So all 3^n states are hyperbolic. If $w \in \{L, H\}^n$, the state is a hyperbolic sink, hence asymptotically stable §2.9 [11]. Otherwise it has a positive real eigenvalue and is unstable.

The explicit family. The parameters of Theorem 3(2) give, through Lemma 12,

$$K = \frac{1}{4n}, \quad \lambda = \frac{1}{200m}, \quad z_c = \frac{1}{20}, \quad \Delta = 6400\, n\, m, \quad X = \lambda\Delta K = 8.$$

(For instance $z_c = \varepsilon + \varepsilon^2\eta_k = \varepsilon + (1-20\varepsilon)/20$.) Take $a = 3/20$, $b = 3$ and $\zeta_+ = \max\{\varepsilon, 9\}$. Then $\lambda \le 1/200$, $1 + K \le 5/4$, and $\lambda\zeta_+ = \max\{\varepsilon, 9\}/(200\max\{1, \varepsilon\}) \le 9/200$. Also $(1+K)^n \le e^{1/4}$, because $n\log\left(1 + \frac{1}{4n}\right) \le \frac{1}{4}$. The hypotheses then follow from five numerical facts, independent of n and ε (`check_construction_inequalities.py`, ball arithmetic):

$$\zeta_+ \;\ge\; 9 \;\ge\; \frac{1/20 + 8}{1 - 1/200} = 8.0905\ldots \;\ge\; \frac{z_c + X/(1+K)}{1-\lambda},$$

$$\text{(W1)}\quad \lambda\zeta_+ + z_c + Xe^{-1/a} \le \tfrac{9}{200} + \tfrac{1}{20} + 8e^{-20/3} = 0.10518\ldots < \tfrac{3}{20},$$

$$\text{(W2)}\quad Xc_-e^{-1/b} \ge \frac{8e^{-1/3}}{(1+K)^{n-1}},$$

$$8e^{-1/3} = 5.7322\ldots > 3e^{1/4} = 3.8520\ldots \ge (1+K)^n\, b,$$

$$\text{(W3)}\quad (1+K)X\psi(\tfrac{3}{20}) \le \tfrac{5}{4}\cdot 8 \cdot \tfrac{400}{9}\, e^{-20/3} = 0.5656\ldots < 1,$$

$$(1+K)X\psi(3) \le \tfrac{5}{4}\cdot\tfrac{8}{9}\, e^{-1/3} = 0.7961\ldots < 1.$$

For (W2), multiply the second inequality by $(1+K)^{-(n-1)}$ and combine it with the first. □

Remark 15 (what the states look like)**.** In the family of Theorem 3(2), a reactor in state L runs below $3E/(20R)$ and converts less than $Ke^{-20/3} < 10^{-3}/n$ of its inlet reactant. In state M it runs between $3E/(20R)$ and $3E/R$. In state H it runs between $3E/R$

and $9E/R$. In every state it converts less than $K/(1+K) = 1/(4n+1)$ of what it receives, so the whole cascade converts less than $1 - e^{-1/4} \approx 22\%$ of the feed. For $n = 3$, $\varepsilon = 1/20$, the certificate gives, for example, $\theta_1 \approx 6.6 \times 10^{-6}$ ($T \approx T_0$) in state L and $\theta_1 \approx 2557.5$ ($T = T_0(1+\varepsilon\theta_1) \approx 129\,T_0$) in state H. These parameters are a mathematical construction, far from any practical operating range. Two features carry it. First, the Arrhenius factor is bounded ($k(T) \le k_0$), so a small $k_0 V/F$ caps the conversion of an ignited reactor; the S-shape of the heat release then comes from the inflection of $e^{-E/RT}$, which lies at $T = E/(2R)$. Second, strong cooling (λ small) makes each reactor almost insensitive to the temperature of its feed. ⌟

Remark 16 (the exponential limit $\varepsilon = 0$)**.** The construction needs $\varepsilon > 0$. With $E(\theta) = e^{\theta}$, an ignited reactor leaves unconverted only the fraction $1/(1+D_k e^{\theta_k})$ of its feed, which is not bounded below as θ_k grows; on the upper branch it is even below one half ($g_k > 1/2$ where $g_k'' < 0$). So downstream reactors receive very different amounts of reactant depending on which upstream reactors are ignited. Theorem 9 holds for $\varepsilon = 0$. We do not know whether 3^n is attained for $\varepsilon = 0$ and $n \ge 3$. The thesis's numerical example for $n = 2$ has $\varepsilon = 0$, but its conjecture concerns the model with $\varepsilon = RT_0/E$, which is positive. ⌟

Remark 17 (identical tanks)**.** The family uses identical tanks. For two identical tanks with $\varepsilon = 0$, Svoronos, Aris and Stephanopoulos [7] report at most seven steady states (for a given Damköhler number), at most three stable. For $\varepsilon > 0$ and $n = 2$, Theorem 14 gives nine, four of them stable, already with identical tanks. ⌟

4 Computer verification

Environment: Fedora 44 under WSL, Python 3.12, SymPy 1.14, python-flint 0.9 (Arb ball arithmetic). From the folder of this problem:

script	claims (PASS/FAIL, exit code)	time
`check_model_algebra.py`	SymPy identities: reduced variables (Lemmas 6 and 12), g'' and G'' factorisations (Lemma 7), det and trace of J_k (Lemma 8), block triangular structure for $n = 3$, ψ'	30 s
`check_construction_inequalities.py`	the numerical facts in the proof of Theorem 14 (Arb), the reductions on samples, thesis parameters for $n = 3, 4$	3 s
`certify_cascades.py`	for $\varepsilon = 1/20$, $n \le 6$, and $\varepsilon \in \{1, 5\}$, $n \le 4$: 3^n steady states isolated reactor by reactor, each verified by a Krawczyk test on (1) in the variables (y_k, θ_k), boxes pairwise disjoint, block traces and determinants enclosed; 2^n stable	15 s
`certify_unstable_outer_state.py`	Proposition 11	2 s

The certificate for $n = 3$ and $n = 4$ with $\varepsilon = 1/20$ lists every steady state (word, enclosures of y_k, θ_k, box radius, traces and determinants of the diagonal blocks) in `certificates/cascade_n3_eps1_20.json` and `cascade_n4_eps1_20.json`. With $\varepsilon = 1/20$, the thesis parameters are $B = 7\,680\,000$ ($n = 3$) and $10\,240\,000$ ($n = 4$), $\mathrm{Da}_0 = e^{-20}/(4n)$, $\tau = 1$, $\alpha_k = 1$, $\beta_k = 199$, $\eta_k = 0$. The certificates illustrate Theorem 14; its proof does not depend on them. The Krawczyk test works on the equations of the thesis, not on the reduced variables, so it also checks the reduction independently.

Limitations. The upper bounds and the construction are proved for the model (1) as written in the thesis: first-order kinetics, equal heat capacities, no wall or jacket dynamics. The attaining parameters are extreme (ignited states above $3E/R$, adiabatic

rise $6400\,n\,m\,E/R$). Whether 3^n states occur within realistic ranges (bounded B, small ε, temperatures below E/R) is not addressed, nor is the case $\varepsilon = 0$. Varma [5], Kubíček et al. [8], Retzloff et al. [9] and Li [10] were checked through abstracts and citing works only.

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